

An Phan

📍 Boca Raton, FL 33486 | ☎ (561) 774-5866 | ✉ tphan2025@fau.edu | 🏠 anphannt.github.io | 📄 an-phan-00b29633a

Education

FAU (College of Engineering & Computer Science - Florida Atlantic University)

FL, U.S

PH.D CANDIDACY IN COMPUTER ENGINEERING

Jan. 2025 - Present

- Advisor: Prof. Xingquan Zhu
- Research area: Machine Learning, Deep Learning, Graph Neural Network (GNN)

HCMUS (University of Science - Vietnam National University)

HCMC, Vietnam

M.S. IN MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Jan. 2021 - Jun. 2023

HCMUE (University of Education)

HCMC, Vietnam

B.S. IN MATHEMATICS

Sep. 2015 - Jun. 2019

Research Experience

Florida Atlantic University (FAU)

FL, U.S

GRADUATE RESEARCH & TEACHING ASSISTANT

Jan. 2025 - Present

- Conduct research in **Machine Learning**, **Deep Learning**, and **Graph Neural Networks (GNNs)** for **healthcare data** analysis, with a focus on real-world clinical and biomedical applications.
- Teach laboratory sections and provided instructional support for **Introduction to Python Programming (COP3035C)**, assisting students with programming fundamentals, debugging, hands-on lab exercises, and grading homework and examinations with constructive feedback.
- Present research findings through a **poster presentation** at the BIOFLORIDA Regional Event and an **oral presentation** at the ICMLA conference, engaging with academic and industry audiences.

AISIA Research Lab

HCMC, Vietnam

RESEARCH ASSISTANT

Sep. 2022 - Jun. 2024

- Led end-to-end project coordination as both a **Research Assistant and Project Manager**, including requirement gathering, timeline planning, technical documentation, and communication between technical and industry partners.
- Explored, cleaned, and standardized large-scale datasets to create **production-ready data pipelines**, enabling downstream machine learning, modeling, and analytics tasks.
- Applied **machine learning, statistical analysis, and data preprocessing** techniques to support prediction, classification, and trend-identification efforts across industry-focused projects.
- Built interactive **analytics dashboards (Tableau)** to support data-driven decision-making for business stakeholders.

Research & Technical Projects

EEG-Based Alzheimer's Disease Detection

FL, U.S

PHD RESEARCH PROJECT

Jan 2025 - Present

- Conducted research under the supervision of my advisor, developing deep learning and graph neural network models for Alzheimer's disease detection using multi-channel EEG signals.
- Extracted and analyzed spectral and connectivity-based features (e.g., PSD, relative band power, coherence) and explored graph-based representations for classification, evaluating performance at both segment and subject levels.
- Designed and trained models using **PyTorch**, focusing on efficient training and scalable experimentation.

AI-Powered Multilingual Self Check-In System for Healthcare

FL, U.S

SOFTWARE ENGINEERING COURSE PROJECT (FAU)

Sep - Dec. 2025

- **Project Overview:** Designed and developed a secure, cloud-based patient self-check-in system integrating AI-powered facial recognition and automated appointment workflows, with a multilingual, patient-friendly interface to improve accessibility, privacy, and efficiency.
- **Role & Contributions:** Designed system logic, functional requirements, and AI components, integrating **pretrained Deep-Face/FaceNet models** for facial recognition. Collaborated closely with front-end and back-end teams to embed AI features into the web application, providing technical feedback and support to ensure accuracy, security, and usability across the system.

Building an Analytics Platform for Reference-Based Pricing in Healthcare

HCMC, Vietnam

AISIA RESEARCH LAB

Mar - May. 2024

- **Developed an analytics platform** to support healthcare decision-making using Reference-Based Pricing benchmarks derived from CMS and payer datasets.
- Designed a **PostgreSQL data warehouse** with dimension and fact tables; performed data cleaning, standardization, and relational schema design to ensure high-quality data pipelines.
- Built **Tableau dashboards** to visualize cost variations, and provide actionable insights for healthcare stakeholders.

Mining and Analyzing Customer Reviews for Hotels in Vietnam

HCMC, Vietnam

AISIA RESEARCH LAB

Oct - Dec. 2023

- Served as **project coordinator** between **AISIA Research Lab** and industry clients, managing requirement gathering, cross-team communication, and project delivery for hospitality analytics projects.
- Cleaned, standardized, and preprocessed large-scale customer review datasets from **Booking.com** and **TripAdvisor** to support downstream analysis.
- Conducted **statistical analysis** and **developed visual dashboards** to identify customer behavior patterns and booking trends in Vietnamese hospitality industry.

Data for Change: Climate Data Analysis Education

Online

COLLABORATIVE PROJECT

Aug. 2023

- **Designed and delivered instructional content** including lesson plans, lecture notes, recorded videos, quizzes, and assignments hosted on Canvas LMS.
- Developed hands-on tutorials guiding learners to acquire, process, and analyze climate datasets using **RStudio**.
- Co-authored the educational eBook “Climate Data Analysis for Beginners,” focusing on accessible, beginner-friendly data analysis workflows.

Stock Portfolio Optimization using Deep Learning

HCMC, Vietnam

MASTER THESIS

Sep. 2022 - Jun. 2023

- Developed a deep learning framework for **portfolio optimization** using historical data from 50 Vietnamese stocks (2016–2021).
- Constructed **3D financial tensors** integrating price data, trading volume, and technical indicators including exponential moving averages (EMA).
- Designed and trained a **BiGRU model with self-attention** implemented in **Keras (TensorFlow)**, achieving improved predictive performance and portfolio stability compared to baseline investment strategies.

Selected Publications (2 of 5)

EDLAD: Ensemble Deep Learning for Alzheimer’s Disease Detection using Brain EEG Signals

ICMLA

AN PHAN, AND XINGQUAN ZHU

Dec. 2025

Portfolio Optimization for Long-Term Investment: An Empirical Study

SoMeT

AN PHAN, NU DINH, SON HUYNH, AND BINH NGUYEN

Sep. 2023

Skills

TECHNICAL

Programming	Python, R, SQL, Bash
ML Frameworks	PyTorch, TensorFlow, scikit-learn, Keras
Data Tools	Pandas, NumPy, Matplotlib, Tableau, PowerBI, Dbeaver, PostgreSQL
AI Areas	Machine Learning, Deep Learning (CNN, LSTM, Transformer), Graph Neural Network (GNNs)
Other	Git, Linux

LANGUAGE

English	Professional working proficiency
Vietnamese	Native or bilingual proficiency